

while the Consumer Price Index (CPI), a proxy for inflation, was up by 0.1 percent. Second, neither TIPS nor I-bonds escape taxation. Both the interest payments and the inflation adjustment gain are eventually taxed as ordinary interest income. More information on inflation-protected securities can be found in [Chapter 8](#).

Risk-free investing is a myth. It does not exist. If there were a risk-free investment, it would have a federal government guarantee, stable pricing, and inflation protection, and the interest income would be free from all city, state, and federal income tax. That being said, if the government guarantees a bond, it must be AAA rated and stay AAA rated, which is a risk in itself. As of this writing, there is no such investment.

DEFINING INVESTMENT RISK

There are many definitions of investment risk, depending on who you are and what you are measuring. Risk can be price volatility, or the dollar amount loss on an investment, or a loss relative to inflation and taxes, or not meeting one's long-term financial objectives.

Academics often define risk as the volatility of price or return over a specified period. Volatility can be measured in different ways and over different periods. You can use price highs or lows or closing prices. You can measure volatility daily, weekly, monthly, or annually. High volatility means erratic investment returns, whereas low volatility means more consistent investment returns. Annual return volatility measurement tends to be the one most commonly used to compare asset class risk.

My view of price volatility is somewhat different from that of the academics. Price and return volatility do not define risk. Rather, they are a derivative of some other risk factor. With stocks, price volatility is based on wavering expectations of earnings and dividend growth. With bonds, it is the wavering expectations of future inflation and interest rates. With commodities, price volatility is caused by changes in the supply and demand curve.

Price volatility is not itself a part of an investment's income stream, and investors should not expect to make excess returns simply because an investment has price volatility. For example, the volatility of commodities is about the same volatility as stocks. However, commodities pay no interest, have no earnings, and pay no dividends. Consequently, the